Model Monitoring Document Guide

1. Purpose of This Guide

This guide explains how to complete, maintain, and use a Model Monitoring Document for AI systems after they are deployed.

A Model Monitoring Document is not just a technical artifact. It is an operational governance document that shows how an organization monitors model performance, safety, fairness, privacy, security, reliability, business value, and continued fitness for use.

The goal of this guide is to help business owners, model owners, technical teams, risk teams, compliance teams, privacy teams, security teams, and auditors understand what should be monitored, why it matters, who owns it, what evidence should be collected, and what action should be taken when problems are found.

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2. What Is a Model Monitoring Document?

A Model Monitoring Document defines the controls, metrics, thresholds, owners, review process, and escalation path used to monitor an AI model or AI system after deployment.

It answers questions such as:

- Is the model still performing as expected?
- Is the model still being used for its approved purpose?
- Has the input data changed?
- Has model performance degraded?
- Are outputs becoming inaccurate, unsafe, biased, or unreliable?
- Are users relying on the model correctly?
- Are human reviewers overriding the model more often?
- Are there privacy or security issues?
- Are there signs of misuse or unauthorized access?
- Should the model be retrained, recalibrated, restricted, or retired?

The document should be created before production deployment and maintained throughout the AI system lifecycle.

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3. When This Document Should Be Created

A Model Monitoring Document should be created before an AI system goes live.

It should be required for:

- Predictive models
- Classification models
- Regression models
- Recommendation systems
- Generative AI systems
- Large language model applications
- Retrieval-augmented generation systems
- AI agents
- Autonomous workflows
- Computer vision systems
- Natural language processing systems
- Third-party AI tools used in business workflows
- AI systems that influence customer, employee, financial, operational, legal, or safety outcomes

For low-risk AI systems, the document may be lightweight.

For moderate-risk and high-risk AI systems, the document should be more detailed and include evidence of monitoring, escalation, human oversight, and periodic review.

4. Who Should Use This Guide?

This guide is intended for the following groups.

Business Owners

Business owners use this guide to confirm that the AI system continues to deliver value and remains aligned with the approved business purpose.

They are responsible for answering:

- Is the model still useful?
- Is the model helping the business?
- Are users using it correctly?
- Are there business risks or customer concerns?
- Should the model continue, change, or be retired?

Model Owners

Model owners use this guide to manage the model's health, performance, risk, documentation, and lifecycle.

They are responsible for answering:

- Is the model performing within approved limits?
- Are monitoring dashboards active?
- Are issues being reviewed?
- Are retraining or recalibration actions needed?
- Are governance reviews complete?

Technical Owners

Technical owners use this guide to implement monitoring pipelines, dashboards, logs, alerts, and technical controls.

They are responsible for answering:

- Are the monitoring systems working?
- Are logs complete?
- Are alerts configured?
- Are data pipelines stable?
- Are model performance metrics accurate?
- Are technical incidents being investigated?

Risk and Compliance Teams

Risk and compliance teams use this guide to verify that controls are operating as expected.

They are responsible for answering:

- Are monitoring requirements documented?
- Are review records available?
- Are high-risk issues escalated?
- Are exceptions approved?
- Are regulatory and policy obligations being met?

Privacy and Security Teams

Privacy and security teams use this guide to review whether the model creates data exposure, unauthorized access, misuse, or adversarial risk.

They are responsible for answering:

- Is sensitive data protected?

- Are prompts and outputs monitored where appropriate?
- Are security events detected?
- Are AI-specific threats being reviewed?
- Are incident response procedures ready?

Internal Audit

Internal audit uses this guide to evaluate whether the model is governed, monitored, documented, and controlled.

They are responsible for answering:

- Is there evidence that monitoring is happening?
- Are issues tracked and resolved?
- Are responsibilities clear?
- Are approvals documented?
- Are reviews performed on schedule?

5. How to Use the Model Monitoring Document

Use the Model Monitoring Document in five stages.

Stage 1: Before Deployment

Before deployment, complete the monitoring plan.

Define:

- Model owner
- Technical owner
- Business owner
- Model purpose
- Approved use case
- Risk level
- Required metrics
- Thresholds
- Monitoring frequency
- Alerting rules
- Human oversight process
- Escalation path

- Incident response process
- Retraining triggers
- Evidence requirements

The model should not move to production until monitoring expectations are defined.

Stage 2: During Production Operation

Once the model is live, use the document to guide ongoing monitoring.

Track:

- Input data quality
- Output quality
- Performance
- Drift
- Fairness
- Security events
- Privacy events
- Human overrides
- User feedback
- Incidents
- Cost
- Business value

Stage 3: During Periodic Reviews

Use the document during monthly, quarterly, or annual model review meetings.

Review:

- Whether the model is still used for the approved purpose
- Whether performance remains acceptable
- Whether drift has occurred
- Whether incidents or complaints have increased
- Whether human oversight is working
- Whether retraining or recalibration is needed
- Whether the model should continue, change, or be retired

Stage 4: During Incidents

Use the document to guide response when something goes wrong.

Examples include:

- Model performance collapse
- Biased output
- Sensitive data leakage
- Security issue
- Unauthorized model use
- Harmful generative AI response
- AI agent taking an unauthorized action
- Customer complaint or regulatory issue

Stage 5: During Audit or Governance Review

Use the document to show evidence that the AI system is actively monitored and controlled.

Evidence may include:

- Dashboard screenshots
- Alert logs
- Model review notes
- Incident reports
- Human review records
- Override records
- Drift reports
- Fairness reports
- Security logs
- Privacy assessments
- Approval records
- Change records

6. Section-by-Section Completion Guide

Section 1: Document Information

This section captures basic administrative details.

Include:

- AI system name
- Model name
- Model version
- Business owner
- Technical owner
- Model owner

- Risk classification
- Deployment environment
- Date deployed
- Monitoring start date
- Review frequency
- Last review date
- Next review date
- Approval authority

Why This Matters

This section creates accountability. Every monitored AI system must have named owners and a clear review schedule.

Good Example

Model Name: Customer Support Response Classifier
Model Owner: AI Operations Team
Business Owner: Customer Experience
Risk Classification: Moderate
Review Frequency: Monthly
Next Review Date: July 15, 2026

Poor Example

Owner: Data Team
Risk Level: Unknown
Review Frequency: As needed

This is not sufficient because ownership and timing are unclear.

7. Section 2: Purpose

This section explains why monitoring is needed for this specific model.

Describe:

- What the model does
- Why the model is important
- What risks monitoring is intended to detect
- What business process the model supports
- What could go wrong if the model is not monitored

Recommended Language

The purpose of monitoring this model is to ensure that it continues to perform as expected, remains aligned with its approved business use, does not produce unacceptable errors or biased outcomes, and does not create security, privacy, compliance, or operational risk.

What to Avoid

Avoid vague statements such as:

- “To monitor the model.”
- “To check performance.”
- “To ensure it works.”

The purpose should explain what monitoring protects against.

8. Section 3: Scope

This section explains what is included and excluded from monitoring.

Include:

- The model or AI system being monitored
- The production environment
- The business process affected
- User groups
- Input data sources
- Output types
- Connected systems
- Vendor components
- Human review process
- Dashboards and logs included

Example Scope Statement

This monitoring document applies to the Customer Support Response Classifier used by the customer experience team to categorize inbound support messages. Monitoring covers input ticket data, classification outputs, confidence scores, reviewer overrides, latency, error rates, customer complaints, and monthly performance reports.

Out-of-Scope Example

This document does not cover unrelated customer support analytics dashboards or non-AI routing rules unless they directly affect model inputs or outputs.

9. Section 4: Model Overview

This section gives reviewers enough context to understand the system.

Include:

- Model name
- Model type
- Model provider
- Model version
- Hosting environment
- Business function
- Primary users
- Affected stakeholders
- Risk classification
- Approved use case
- Prohibited use cases

Guidance by Model Type

For a predictive model, describe what it predicts.

For a classification model, describe what categories it assigns.

For a generative AI system, describe what content it generates.

For a RAG system, describe what knowledge sources it retrieves from.

For an AI agent, describe what tools it can use and what actions it can take.

10. Section 5: Approved Use Case

This is one of the most important sections.

Clearly state what the model is allowed to do.

A strong approved use case includes:

- Business process
- User group
- Input data
- Output type
- Decision role
- Human oversight requirement
- Limitations

Strong Example

The model is approved to classify inbound customer support tickets into predefined categories to assist support team routing. The model does not make final customer decisions, does not send customer responses directly, and must be reviewed by a support agent when the confidence score is below the approved threshold.

Weak Example

The model helps customer support.

This is too vague and does not define boundaries.

11. Section 6: Prohibited Uses

This section defines what the model must not be used for.

Include prohibited uses such as:

- Use outside approved business purpose
- Final high-impact decisions without human review
- Processing unauthorized sensitive data
- External customer communication without approval
- Legal, medical, financial, or HR decisions without qualified review
- Autonomous actions outside approved boundaries
- Use by unauthorized users
- Use in unapproved systems

Why This Matters

Many AI risks come from model misuse or use-case expansion. Clear prohibited-use language prevents teams from quietly repurposing a model without governance review.

12. Section 7: Monitoring Ownership

This section defines who is accountable for monitoring.

At minimum, define:

- Business owner
- Model owner
- Technical owner
- Monitoring owner
- Compliance reviewer
- Security reviewer
- Privacy reviewer
- Human reviewer group

Recommended Responsibility Matrix

Role	Main Responsibility
Business Owner	Confirms business value and approved use
Model Owner	Owns model performance and lifecycle
Technical Owner	Owns monitoring infrastructure and logs
Risk / Compliance	Reviews governance evidence
Security	Reviews security events and threats
Privacy	Reviews personal data and sensitive data risks
Human Reviewers	Review, approve, reject, or escalate outputs

Common Mistake

Do not assign ownership only to a general team name. A named role or accountable group should be listed.

13. Section 8: Monitoring Frequency

Monitoring frequency should match risk.

Use this guidance:

Risk Level	Recommended Monitoring	Recommended Review
Low Risk	Monthly or quarterly	Quarterly or annually

Risk Level	Recommended Monitoring	Recommended Review
Moderate Risk	Weekly or monthly	Monthly or quarterly
High Risk	Daily, weekly, or continuous	Monthly or quarterly
Critical / Restricted	Continuous or near real-time	Weekly or monthly

Factors That Increase Monitoring Frequency

Increase monitoring frequency when the model:

- Affects customers
- Affects employees
- Affects financial outcomes
- Affects legal or regulatory outcomes
- Uses sensitive data
- Produces external-facing content
- Takes autonomous actions
- Has high usage volume
- Has a history of incidents
- Is newly deployed
- Has recently changed

14. Section 9: Input Data Monitoring

Input data monitoring checks whether the model is receiving the right data.

Monitor:

- Missing values
- Invalid values
- Unexpected fields
- Schema changes
- Data freshness
- Data source availability
- Duplicate records
- Input volume spikes
- Outlier rates
- Unauthorized sensitive data

Why This Matters

A model can fail even if the model itself has not changed. If input data changes, breaks, becomes stale, or includes unauthorized data, model outputs may become unreliable or risky.

Evidence to Collect

- Data quality reports
- Schema validation logs
- Data pipeline alerts
- Input distribution charts
- Data freshness reports
- Data source availability logs

Example Thresholds

Metric	Warning Threshold	Critical Threshold
Missing required fields	More than 2%	More than 5%
Data freshness delay	More than 24 hours	More than 48 hours
Unexpected schema change	Any change	Any production-breaking change
Unauthorized sensitive data	Any occurrence	Any occurrence

15. Section 10: Output Quality Monitoring

Output quality monitoring checks whether the model is producing acceptable results.

For predictive models, monitor:

- Accuracy
- Precision
- Recall
- F1 score
- False positive rate
- False negative rate
- Confidence distribution
- Error rate

For generative AI systems, monitor:

- Hallucination rate
- Groundedness
- Citation accuracy
- Toxicity
- Sensitive data exposure
- Brand alignment
- Refusal accuracy
- Human edit rate

For AI agents, monitor:

- Task completion rate
- Tool call success rate
- Unauthorized action attempts
- Human approval rate
- Human rejection rate
- Escalation rate
- Cost per task

Evidence to Collect

- Model performance reports
- Human review samples
- QA reports
- Error analysis
- Output audit logs
- Customer complaint records
- Human override records

16. Section 11: Drift Monitoring

Drift monitoring detects whether the model's operating environment has changed.

Data Drift

Data drift means the input data distribution has changed.

Examples:

- Customer behavior changes
- New product categories
- New user segments
- New geographies
- New data collection method
- Upstream system change

Model Drift

Model drift means model performance has declined.

Examples:

- Lower accuracy
- Higher false positive rate
- Higher false negative rate
- More human overrides
- More complaints
- Lower business value

Concept Drift

Concept drift means the relationship between inputs and outcomes has changed.

Examples:

- Fraud patterns change
- Customer expectations change
- Market conditions change
- Business policy changes
- Language patterns change

Evidence to Collect

- Drift reports
- Feature distribution comparisons
- Performance trend charts
- Human override trend reports
- Business outcome trend reports

Common Drift Metrics

- Population Stability Index
- Kolmogorov-Smirnov test
- Jensen-Shannon divergence
- Feature mean shift
- Feature variance shift
- Category frequency shift
- Missingness shift

17. Section 12: Fairness and Bias Monitoring

Fairness monitoring is required when the model affects people.

This includes models used in:

- Hiring
- Promotion
- Performance management
- Lending
- Insurance
- Healthcare
- Education
- Fraud detection
- Customer eligibility
- Pricing
- Access to services
- Employee monitoring

What to Monitor

- Performance across relevant groups
- Error rates across groups
- False positives across groups
- False negatives across groups
- Approval or denial rates across groups
- Complaint rates across groups
- Human override rates across groups

Evidence to Collect

- Fairness assessment reports
- Subgroup performance reports
- Bias testing results
- Complaint analysis
- Human review records
- Remediation records

Important Guidance

Fairness monitoring must be designed carefully. Not every system should collect or process sensitive demographic data. Privacy, legal, and compliance teams should approve the fairness monitoring approach.

18. Section 13: Explainability Monitoring

Explainability monitoring checks whether users and reviewers can understand model outputs well enough to use them responsibly.

Monitor whether:

- Explanations are available
- Explanations are understandable
- Human reviewers can use explanations
- Explanations match model behavior
- Users are not misinterpreting outputs
- Explanation tools are working
- High-impact decisions include adequate rationale

Evidence to Collect

- Explanation samples
- Reviewer feedback
- User training records
- Audit records
- Appeal or review records
- Model card or system card

Strong Practice

For high-risk models, explanations should support human review, audit, challenge, appeal, and root cause analysis.

19. Section 14: Generative AI Monitoring

Generative AI requires special monitoring because outputs may be fluent but inaccurate, unsafe, biased, or unsupported.

Monitor:

- Hallucinations
- Unsupported claims
- Fabricated citations
- Toxicity
- Sensitive data exposure
- Copyright concerns
- Brand misalignment
- Policy violations
- Refusal failures
- Customer complaints
- Human edit rates

Evidence to Collect

- Prompt and response samples where allowed
- Red-team test results
- Safety evaluation results
- Content moderation logs
- Citation checks
- Human review records
- Customer feedback
- Incident reports

Review Questions

Ask:

1. Is the system making unsupported claims?
2. Is the system exposing confidential information?
3. Is the system refusing prohibited requests correctly?
4. Is the system producing harmful or biased responses?
5. Is the system being used outside its approved scope?
6. Are users over-trusting generated content?

20. Section 15: RAG System Monitoring

RAG systems must be monitored at both the retrieval layer and the generation layer.

Retrieval Layer Monitoring

Monitor:

- Retrieval relevance
- Retrieval accuracy
- Source freshness
- Missing source rate
- Stale document retrieval
- Unauthorized document retrieval
- Knowledge base coverage
- Access control enforcement

Generation Layer Monitoring

Monitor:

- Groundedness
- Citation accuracy
- Answer completeness
- Hallucination rate
- Unsupported claims
- Over-answering when sources are weak

Evidence to Collect

- Retrieval logs
- Source documents used
- Citation validation samples
- Knowledge base update logs
- Access control test results
- User feedback
- Evaluation reports

Strong Practice

A RAG system should not answer confidently when retrieved evidence is weak, outdated, or unavailable.

21. Section 16: AI Agent Monitoring

AI agents need stronger monitoring than passive AI systems because they can take actions.

Monitor:

- Tool calls
- API calls
- External communications
- File access
- Database access
- System modifications
- Task completion rate
- Failed actions
- Unauthorized action attempts
- Human approval rate
- Human rejection rate
- Looping behavior
- Cost per task
- Escalation rate

Required Agent Controls

AI agents should have:

- Clear task boundaries
- Approved tool list
- Least privilege access
- Human approval for sensitive actions
- Logging of all actions
- Rate limits
- Cost limits
- Escalation path
- Kill switch
- Rollback plan

Evidence to Collect

- Agent action logs
- Tool call logs
- Approval records
- Rejection records
- Cost reports
- Error logs
- Incident reports

Escalate Immediately If

- The agent attempts unauthorized access
- The agent performs an unapproved action
- The agent sends unauthorized external communication
- The agent modifies data incorrectly
- The agent loops repeatedly
- The agent bypasses human approval
- The agent accesses sensitive data unexpectedly

22. Section 17: Security Monitoring

Security monitoring detects misuse, abuse, unauthorized access, and AI-specific attacks.

Monitor:

- Unauthorized access attempts
- Suspicious usage patterns

- Prompt injection attempts
- Jailbreak attempts
- Sensitive data in prompts
- Sensitive data in outputs
- Model extraction attempts
- Abnormal API usage
- Tool misuse
- System prompt leakage
- Malicious file uploads
- Adversarial inputs
- Credential exposure

Evidence to Collect

- Security logs
- Access logs
- Prompt attack reports
- Abuse monitoring reports
- API usage reports
- Incident tickets
- Security investigation records

Strong Practice

For LLM applications, security teams should treat prompts, retrieved documents, tool outputs, and user-uploaded files as potentially hostile inputs.

23. Section 18: Privacy Monitoring

Privacy monitoring ensures that the model does not misuse or expose personal, sensitive, confidential, or regulated data.

Monitor:

- Personal data in prompts
- Sensitive data in prompts
- Personal data in outputs
- Sensitive data in outputs
- Unauthorized profiling
- Unexpected inference of sensitive traits
- Improper prompt or output retention
- Vendor data use changes
- Cross-border data transfer issues

- Individual rights requests
- Privacy complaints

Evidence to Collect

- Privacy review records
- Data flow diagrams
- Prompt/output sampling reports where allowed
- Access logs
- Vendor data processing records
- Data retention evidence
- Deletion request records
- Privacy incident tickets

Escalate Immediately If

- Sensitive data is exposed
- Personal data is processed outside the approved purpose
- Unauthorized users access personal data
- Vendor terms change materially
- Data retention rules are violated
- Regulatory notification may be required

24. Section 19: Operational Monitoring

Operational monitoring checks whether the model service remains available, stable, and efficient.

Monitor:

- Uptime
- Latency
- Throughput
- Error rate
- Timeout rate
- Queue depth
- Dependency failures
- Recovery time
- API availability
- Infrastructure health

Evidence to Collect

- System dashboards
- Availability reports
- Incident reports
- Latency reports
- Error logs
- Service-level reports
- Dependency monitoring records

Example Thresholds

Metric	Warning Threshold	Critical Threshold
Uptime	Below 99.5%	Below 99%
Latency	Above 2 seconds	Above 5 seconds
Error Rate	Above 2%	Above 5%
Timeout Rate	Above 1%	Above 3%

25. Section 20: Cost and Usage Monitoring

Cost and usage monitoring prevents uncontrolled AI spending and detects abnormal behavior.

Monitor:

- Request volume
- Active users
- Token usage
- Compute usage
- Cost per request
- Cost per task
- Monthly spend
- Failed request cost
- Repeated calls
- Agent looping cost
- Vendor price changes

Evidence to Collect

- Billing reports
- Usage dashboards
- Token usage reports
- Compute usage reports
- Cost anomaly alerts
- Vendor invoices
- Budget review records

Escalate If

- Usage spikes unexpectedly
- Monthly spend exceeds budget
- Agent loops create repeated calls
- Failed requests create material cost
- Vendor pricing changes materially
- Cost exceeds business value

26. Section 21: Business Outcome Monitoring

A model can be technically accurate but still fail to deliver business value.

Monitor:

- Productivity improvement
- Cost reduction
- Revenue impact
- Customer satisfaction
- Case resolution rate
- Response time
- Error reduction
- User adoption
- Employee satisfaction
- Operational efficiency

Evidence to Collect

- Business KPI reports
- User adoption reports
- Customer feedback
- Productivity analysis
- Cost-benefit analysis
- Business owner review notes

Review Questions

Ask:

1. Is the model still solving the original problem?
2. Are users adopting the model?
3. Is the model improving business outcomes?
4. Is the model creating unexpected work?

5. Is the cost justified?
6. Should the model be expanded, restricted, replaced, or retired?

27. Section 22: Human Oversight Monitoring

Human oversight monitoring ensures people remain meaningfully accountable.

Monitor:

- Required review completion rate
- Human override rate
- Human rejection rate
- Escalation rate
- Missed review rate
- Review time
- Reviewer disagreement rate
- Blind acceptance patterns

Evidence to Collect

- Human review logs
- Approval records
- Override records
- Rejection records
- Escalation tickets
- Reviewer training records
- Quality assurance reports

Signs of Weak Oversight

- Reviewers approve almost everything without changes
- Reviewers do not understand model limitations
- Overrides are not documented
- Escalations are ignored
- Required reviews are skipped
- AI outputs are treated as final decisions without human judgment

28. Section 23: User Feedback and Complaints

User feedback helps detect quality, trust, fairness, and usability problems.

Collect feedback from:

- End users
- Customers
- Employees
- Human reviewers
- Customer support teams
- Quality assurance teams
- Compliance teams
- Help desk tickets
- Surveys
- Complaints
- Incident reports

Monitor

- User satisfaction
- Complaint volume
- Complaint severity
- Incorrect output reports
- Harmful output reports
- Trust score
- Feature requests
- Repeated failure themes

Evidence to Collect

- Feedback summaries
- Complaint reports
- Survey results
- Help desk tickets
- QA review notes
- Remediation records

29. Section 24: Alerting and Escalation

This section defines what happens when monitoring detects a problem.

Use severity levels.

Low Severity

Minor issue with limited impact.

Examples:

- Small latency increase
- Minor internal output error
- Temporary data delay

Action:

- Track
- Review during next monitoring cycle
- Fix if recurring

Medium Severity

Issue affects quality, user trust, or operations.

Examples:

- Accuracy decline
- Increased human overrides
- Repeated incorrect outputs
- Moderate error rate increase

Action:

- Investigate
- Assign owner
- Define remediation
- Track to closure

High Severity

Issue may affect individuals, compliance, privacy, security, or customer trust.

Examples:

- Biased output
- Sensitive data exposure
- Unauthorized model use
- Harmful customer-facing response
- High-risk model performance failure

Action:

- Escalate immediately
- Notify governance, legal, privacy, security, or compliance as needed
- Restrict or pause use if necessary

Critical Severity

Severe harm, regulatory exposure, security compromise, or major operational disruption.

Examples:

- Major data leakage
- AI agent takes unauthorized production action
- Unlawful or discriminatory decision support
- Major outage affecting critical operations

Action:

- Activate incident response
- Disable or restrict the system if needed
- Preserve evidence
- Notify required stakeholders
- Complete root cause analysis

30. Section 25: Incident Response

This section explains how AI incidents are handled.

An AI incident may involve:

- Incorrect high-impact recommendation
- Biased or discriminatory output
- Sensitive data leakage
- Unauthorized access
- Model producing harmful content
- AI agent taking unauthorized action
- Excessive hallucination
- Model performance collapse
- Model used outside approved scope
- Vendor model change causing risk

Incident Response Steps

1. Record the incident.

2. Determine severity.
3. Preserve logs and evidence.
4. Notify required owners.
5. Restrict, pause, or disable the model if needed.
6. Identify affected users, customers, systems, and data.
7. Investigate root cause.
8. Apply corrective action.
9. Notify affected parties or regulators where required.
10. Document resolution.
11. Update monitoring controls.
12. Review lessons learned.

Evidence to Collect

- Incident report
- Timeline
- Logs
- Screenshots
- Affected records
- Root cause analysis
- Corrective action plan
- Approval to restart if paused
- Lessons learned

31. Section 26: Retraining, Recalibration, and Replacement

Monitoring should define when the model must be retrained, recalibrated, restricted, replaced, or retired.

Retraining Triggers

Retraining may be needed when:

- Accuracy drops below threshold
- Drift exceeds threshold
- Human override rate increases
- Complaint rate increases
- Business rules change
- New data becomes available
- Bias metrics worsen
- External conditions change

Recalibration Triggers

Recalibration may be needed when:

- Confidence scores no longer match outcomes
- Thresholds are no longer appropriate
- False positives increase
- False negatives increase
- Business tolerance changes

Replacement Triggers

Replacement may be needed when:

- The model cannot meet performance requirements
- Risk cannot be reduced
- Vendor support ends
- Better approved options exist
- Cost is too high
- Regulatory expectations change
- The model repeatedly fails monitoring reviews

32. Section 27: Change Management

Any material change to the model should trigger review.

Material changes include:

- New model version
- New training data
- New data source
- New vendor model
- New system prompt
- New RAG knowledge source
- New agent tool
- New user group
- New customer-facing use
- New autonomy level
- New decision use case
- New risk classification
- New monitoring threshold

Required Change Steps

1. Document the proposed change.
2. Assess impact on risk.
3. Review privacy, security, data, fairness, and compliance impacts.
4. Test the updated model.
5. Update documentation.
6. Obtain required approvals.
7. Communicate changes to users.
8. Monitor closely after deployment.

33. Section 28: Monitoring Evidence and Recordkeeping

Evidence proves that monitoring is actually happening.

Maintain records such as:

- Monitoring dashboard screenshots
- Alert logs
- Drift reports
- Performance reports
- Fairness reports
- Security logs
- Privacy logs
- Incident reports
- Human review records
- Override records
- Complaint records
- Model review meeting notes
- Approval records
- Change records
- Retraining records
- Retirement records

Evidence Quality Standard

Good evidence should show:

- Date
- Model name
- Model version

- Metric or issue reviewed
- Owner
- Decision made
- Action taken
- Status
- Approval if required

34. Section 29: Dashboard Guidance

A model monitoring dashboard should make model health visible.

Recommended dashboard sections:

- Model name
- Model version
- Risk level
- Deployment status
- Input data quality
- Output quality
- Model performance
- Drift
- Bias and fairness
- Security alerts
- Privacy alerts
- Human oversight metrics
- Cost and usage
- User feedback
- Incidents
- Open remediation items
- Last review date
- Next review date

Dashboard Design Guidance

A useful dashboard should:

- Highlight red, yellow, and green status
- Show trends over time
- Show thresholds
- Identify owners
- Show open issues
- Support audit export
- Separate technical metrics from business metrics

- Include high-risk alerts prominently

35. Section 30: Periodic Model Review Meeting

Model review meetings should be held according to the model's risk level.

Recommended Agenda

1. Confirm model is still used for approved purpose.
2. Review model performance.
3. Review input data quality.
4. Review drift indicators.
5. Review fairness and bias indicators.
6. Review explainability issues.
7. Review human oversight metrics.
8. Review security and privacy alerts.
9. Review user feedback and complaints.
10. Review incidents and remediation.
11. Review cost and business value.
12. Decide whether to continue, retrain, restrict, suspend, or retire the model.

Possible Review Decisions

- Continue use
- Continue with remediation
- Increase monitoring
- Increase human oversight
- Retrain model
- Recalibrate model
- Restrict use
- Suspend use
- Replace model
- Retire model
- Escalate to AI Governance Committee

36. Model Monitoring Maturity Levels

Organizations can use maturity levels to assess how strong their monitoring process is.

Level 1: Basic

- Model has an owner
- Basic performance metrics are reviewed
- Issues are handled manually
- Documentation is limited

Level 2: Defined

- Monitoring metrics are documented
- Thresholds are defined
- Reviews happen on a schedule
- Incidents are tracked
- Ownership is clear

Level 3: Managed

- Dashboards are active
- Alerts are automated
- Drift is monitored
- Human overrides are tracked
- Evidence is retained
- Governance reviews occur

Level 4: Advanced

- Monitoring is risk-based
- Fairness and privacy indicators are tracked
- AI-specific security events are monitored
- Retraining triggers are automated or semi-automated
- Audit evidence is consistently available

Level 5: Optimized

- Monitoring is continuous for high-risk systems
- Controls are integrated into AI lifecycle workflows
- Business value and risk are reviewed together
- AI incidents drive control improvements
- Governance, technical, and business teams share a common operating model

37. Common Mistakes to Avoid

Avoid these common mistakes when creating or using a Model Monitoring Document.

Mistake 1: Monitoring Only Accuracy

Accuracy is important, but it is not enough.

Also monitor:

- Drift
- Bias
- Human overrides
- User complaints
- Security events
- Privacy events
- Operational reliability
- Cost
- Business value

Mistake 2: No Thresholds

A metric without a threshold does not tell the team when to act.

Every important metric should have:

- Baseline
- Warning threshold
- Critical threshold
- Owner
- Required action

Mistake 3: No Human Owner

A dashboard does not create accountability by itself.

Every model needs:

- Business owner
- Model owner
- Technical owner
- Escalation owner

Mistake 4: Ignoring Model Use After Deployment

Many AI risks appear after deployment because users apply the model in new ways.

Monitor whether the model is still being used only for its approved purpose.

Mistake 5: Treating Generative AI Like Traditional ML

Generative AI needs additional monitoring for:

- Hallucination
- Toxicity
- Groundedness
- Sensitive data exposure
- Prompt injection
- Citation accuracy
- Policy violations
- Misuse

Mistake 6: Not Monitoring AI Agents

AI agents require action-level monitoring because they can interact with systems and tools.

Monitor:

- What tools they use
- What actions they take
- Whether humans approved sensitive actions
- Whether they attempted unauthorized behavior

Mistake 7: No Evidence Retention

If monitoring is not documented, it is difficult to prove it happened.

Keep records of reviews, alerts, incidents, decisions, and remediation.

38. Practical Completion Checklist

Before finalizing a Model Monitoring Document, confirm:

Item	Complete
Model name and version are documented.	☐
Business owner is assigned.	☐
Model owner is assigned.	☐
Technical owner is assigned.	☐
Risk level is documented.	☐
Approved use case is clear.	☐
Prohibited uses are clear.	☐
Monitoring frequency is defined.	☐
Input data metrics are defined.	☐
Output quality metrics are defined.	☐
Drift metrics are defined.	☐
Fairness metrics are defined where required.	☐
Security monitoring is defined.	☐
Privacy monitoring is defined.	☐
Human oversight metrics are defined.	☐
Business outcome metrics are defined.	☐
Thresholds are defined.	☐
Alerts are defined.	☐
Escalation path is defined.	☐
Incident response process is defined.	☐
Retraining triggers are defined.	☐
Change management process is defined.	☐
Evidence retention requirements are defined.	☐
Review meeting cadence is defined.	☐
Approval is documented.	☐

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39. Example Monitoring Summary

Model Monitoring Summary Example

Model Name: Customer Support Ticket Classifier
Model Version: 2.1
Review Period: June 1–June 30, 2026

Risk Level: Moderate
Prepared By: Model Owner
Overall Status: Yellow

Summary

The model remains in production and continues to support the approved customer support ticket routing use case. Overall classification accuracy remains above the approved threshold. However, the human override rate increased from 8% to 14%, mainly in billing-related tickets. Input data quality remains acceptable, and no privacy or security incidents were identified during this period.

Key Findings

- Accuracy remains within approved limits.
- Override rate increased above warning threshold.
- Billing ticket category shows higher misclassification.
- No unauthorized data access detected.
- No sensitive data exposure detected.
- No customer complaints directly attributed to the model.
- Business value remains positive due to reduced manual triage time.

Actions

Issue	Owner	Action	Due Date
Increased override rate for billing tickets	Model Owner	Review recent billing ticket samples	July 10
Possible category drift	Technical Owner	Compare current billing tickets to training data	July 15
Reviewer feedback needed	Business Owner	Collect examples from support agents	July 12

Decision

Continue model use with remediation. Review billing category performance again during the next monthly monitoring review.

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40. Final Guidance

A Model Monitoring Document should be treated as a living governance document.

It should be updated when:

- The model changes
- Data changes
- Users change
- Business purpose changes
- Risk level changes
- Monitoring thresholds change
- Incidents occur
- New laws or regulations apply
- Vendor terms change
- The model is retrained, replaced, restricted, or retired

A strong Model Monitoring Document helps the organization prove that AI is not only deployed, but actively governed throughout its lifecycle.